Verified experimental comparisons

CLIP ViT-B/32. Accuracy (%). Prepared 22 October 2026. Each result uses its matching unprotected control. Historical tests have been observed. Missing experiments are explicitly marked; no eight-task average is reported.

Table 1: Main comparison: two experts, TA coefficient 0.5

Protection	Standalone M	No recovery M / S	AGA target M / S	AGA both M / S
Unprotected	99.69	99.37 / 92.50	99.30 / 92.78	99.32 / 91.77
PaRaMS (full)	99.69	8.92 / 9.15	99.30 / 92.79	99.32 / 91.77
Defense 1	99.69	9.80 / 6.70	9.80 / 6.70	9.80 / 6.70
O/B [H]	99.71	78.77 / 62.94	79.53 / 65.82	77.85 / 61.42
A/B [H]	99.65	91.07 / 77.46	85.42 / 75.50	79.20 / 69.84
Defense 2 (structure) [K]	99.69	7.39 / 7.74	10.09 / 6.78	10.09 / 6.78
Defense 2 candidate (failed) [F]	99.69	99.13 / 91.41	99.01 / 91.59	99.04 / 90.51
Defense 2 (task-directed)	99.69	10.98 / 14.32	9.80 / 9.81	9.80 / 9.83

M: protected MNIST; S: partner SVHN. AGA target/both denotes recovery of only the protected expert/both experts. [H] Historical O/B and A/B fail the original complete-task bound; effect references only. [K] Known global symmetry control, not an original method; family-aware public recovery is reported separately. [F] Previous candidate preserves the complete task check but does not provide useful protection. Full PaRaMS is the full-default baseline; parameter scales differ and are not a universal exclusion rule.

Table 2: Matched defenses across three protected tasks

Target / partner	Protection	Standalone	No recovery T / P	AGA target T / P	AGA both T / P
MNIST / SVHN	Unprotected	99.69	99.37 / 92.50	99.30 / 92.78	99.32 / 91.77
MNIST / SVHN	Defense 1	99.69	9.80 / 6.70	9.80 / 6.70	9.80 / 6.70
MNIST / SVHN	PaRaMS (full)	99.69	8.92 / 9.15	99.30 / 92.79	99.32 / 91.77
MNIST / SVHN	Defense 2 (structure)	99.69	7.39 / 7.74	10.09 / 6.78	10.09 / 6.78
MNIST / SVHN	Defense 2 (task-directed)	99.69	10.98 / 14.32	9.80 / 9.81	9.80 / 9.83
DTD / MNIST	Unprotected	79.41	69.31 / 99.45	68.40 / 99.45	68.78 / 99.42
DTD / MNIST	Defense 1	79.41	2.13 / 21.09	2.13 / 21.09	2.13 / 21.05
DTD / MNIST	PaRaMS (full)	79.41	2.07 / 8.92	68.35 / 99.45	68.78 / 99.42
DTD / MNIST	Defense 2 (structure)	79.41	2.23 / 8.68	2.18 / 9.89	2.13 / 9.85
DTD / MNIST	Defense 2 (task-directed)	79.41	2.39 / 10.29	2.39 / 10.09	2.39 / 10.09
SVHN / MNIST	Unprotected	97.46	92.50 / 99.37	91.56 / 99.39	91.77 / 99.32
SVHN / MNIST	Defense 1	97.46	6.70 / 9.80	6.70 / 9.80	6.70 / 9.80
SVHN / MNIST	PaRaMS (full)	97.46	9.09 / 8.95	91.56 / 99.39	91.77 / 99.32
SVHN / MNIST	Defense 2 (structure)	97.46	7.74 / 7.39	9.74 / 10.28	9.63 / 10.28
SVHN / MNIST	Defense 2 (task-directed)	97.46	7.30 / 9.45	6.45 / 12.19	6.45 / 12.22

TA0.5 and unchanged original AGA; every merged cell is protected / partner accuracy. Full PaRaMS uses official defaults without clipping. Structure uses a fixed global rule; task-directed uses the frozen finite objective with each target task SVD. Three protected tasks correspond to two unordered task pairs, not a complete eight-task table. Novelty and known-family/location-aware recovery are separate. Complete original-task predictions, parameter scales and numerical protocols are reported separately.

Table 3: Defense 1 across three protected tasks

Target / partner	Protection	Standalone	No recovery T / P	AGA target T / P	AGA both T / P
MNIST / SVHN	Unprotected	99.69	99.37 / 92.50	99.30 / 92.78	99.32 / 91.77
MNIST / SVHN	Defense 1	99.69	9.80 / 6.70	9.80 / 6.70	9.80 / 6.70
MNIST / DTD	Unprotected	99.69	99.45 / 69.31	99.43 / 69.41	99.42 / 68.78
MNIST / DTD	Defense 1	99.69	9.80 / 2.13	9.80 / 2.13	9.80 / 2.13
DTD / MNIST	Unprotected	79.41	69.31 / 99.45	68.40 / 99.45	68.78 / 99.42
DTD / MNIST	Defense 1	79.41	2.13 / 21.09	2.13 / 21.09	2.13 / 21.05
SVHN / MNIST	Unprotected	97.46	92.50 / 99.37	91.56 / 99.39	91.77 / 99.32
SVHN / MNIST	Defense 1	97.46	6.70 / 9.80	6.70 / 9.80	6.70 / 9.80

TA0.5. Every merged cell is target / partner accuracy. The task role is fixed by the first column, not by a global dataset order. The original AGA does not modify the final LayerNorm bias; these results do not imply resistance to location-aware or data-assisted recovery.

Table 4: Recovery permissions: protected MNIST and partner SVHN

Recovery pipeline	Additional access	Matched control M / S	Defense 1 M / S
No recovery	Published expert and partner	99.37 / 92.50	9.80 / 6.70
AGA target	Public base; defense-agnostic	99.30 / 92.78	9.80 / 6.70
AGA both	Public base; defense-agnostic	99.32 / 91.77	9.80 / 6.70
Diagonal adaptation	Specified diagonal family; no images	99.37 / 92.50	9.80 / 6.70
LN-bias cleanup	Known location; published projection	99.37 / 92.50	99.37 / 92.50
Student + TA	2048 target-domain queries; student training	95.53 / 95.95	95.53 / 95.95
Student + AGA target	2048 target-domain queries; student training	95.35 / 95.94	95.35 / 95.94
Student + AGA both	2048 target-domain queries; student training	95.25 / 95.29	95.25 / 95.29

All merges use TA0.5. Diagonal adaptation is the official MLP kernel plus an explicitly specified native Attention adaptation, not a complete official attack release. Cleanup is a known-location diagnostic. Native FP32 cleanup changes some scores beyond the historical numerical-control tolerance; no exact FP32 function identity is claimed. Student rows share one training problem because all2048 queried hard labels coincide; they are not independent seed replications or the complete PaRaMS KD implementation.

Table 5: Supplementary mergers after original AGA on the protected expert

Merger	Unprotected M / S	Defense 1 M / S	O/B [H] M / S	A/B [H] M / S
TA	99.30 / 92.78	9.80 / 6.70	79.53 / 65.82	85.42 / 75.50
TIES	99.32 / 94.81	9.82 / 9.69	57.13 / 50.75	65.76 / 66.21
DARE+TA	99.30 / 92.81	24.20 / 25.50	79.52 / 66.01	86.11 / 75.87
NUFILT M→S	99.39 / 94.68	99.39 / 94.68	99.24 / 93.80	99.14 / 94.61
NUFILT S→M	99.07 / 95.35	99.07 / 95.35	98.97 / 95.07	98.86 / 95.38
ProDistill	99.42 / 96.83	9.80 / 6.70	99.42 / 96.16	99.42 / 96.46

MNIST target / SVHN partner. Each merger has its own matching control. DARE+TA is the mean of fixed mask seeds42/43/44; other entries are fixed runs. NUFILT reports both task orders and uses native adapters of official kernels. ProDistill uses64+64 unlabeled training calibration images and attacker-side elementwise coefficient training; it is not a data-free attack. Mergers are not counted as separate recovery attacks. [H] Task-bound failure retained.

Table 6: DARE mask variability under AGA target

Protection	MNIST mean ± SD	SVHN mean ± SD
Unprotected	99.30 ± 0.00	92.81 ± 0.04
Defense 1	24.20 ± 5.60	25.50 ± 10.30
O/B [H]	79.52 ± 0.57	66.01 ± 0.71
A/B [H]	86.11 ± 0.95	75.87 ± 0.56

Sample standard deviation across the same three fixed masks42/43/44. This is mask variability, not three independent expert or defense trainings. All three mask values are retained in the cited source rows.

Table 7: Supplementary mergers: DTD and SVHN protected experts

Target / partner	Merger / order	AGA scope	Control T / P	Defense 1 T / P
DTD / MNIST	NUFILT DTD / MNIST	None	72.18 / 99.45	72.18 / 99.45
DTD / MNIST	NUFILT DTD / MNIST	Target	70.59 / 99.45	70.59 / 99.45
DTD / MNIST	NUFILT DTD / MNIST	Both	70.90 / 99.42	70.90 / 99.42
DTD / MNIST	NUFILT MNIST / DTD	None	72.71 / 99.51	72.71 / 99.51
DTD / MNIST	NUFILT MNIST / DTD	Target	70.96 / 99.51	70.96 / 99.51
DTD / MNIST	NUFILT MNIST / DTD	Both	71.44 / 99.48	71.44 / 99.48
DTD / MNIST	ProDistill	None	73.14 / 99.51	2.13 / 98.06
DTD / MNIST	ProDistill	Target	73.09 / 99.52	2.13 / 98.07
DTD / MNIST	ProDistill	Both	73.40 / 99.47	2.13 / 97.71
SVHN / MNIST	NUFILT SVHN / MNIST	None	95.11 / 99.19	95.11 / 99.19
SVHN / MNIST	NUFILT SVHN / MNIST	Target	94.40 / 99.25	94.40 / 99.25
SVHN / MNIST	NUFILT SVHN / MNIST	Both	94.51 / 99.14	94.51 / 99.14
SVHN / MNIST	NUFILT MNIST / SVHN	None	94.45 / 99.43	94.45 / 99.43
SVHN / MNIST	NUFILT MNIST / SVHN	Target	93.47 / 99.44	93.47 / 99.44
SVHN / MNIST	NUFILT MNIST / SVHN	Both	93.75 / 99.39	93.75 / 99.39
SVHN / MNIST	ProDistill	None	96.73 / 99.45	6.70 / 9.80
SVHN / MNIST	ProDistill	Target	96.45 / 99.45	6.70 / 9.80
SVHN / MNIST	ProDistill	Both	96.56 / 99.44	6.70 / 9.80

NUFILT uses both task orders. SVHN no-attack/both-AGA cells reuse the identical earlier MNIST/SVHN merger models with dataset roles exchanged; no new fits. The selected-parameter identity for Defense1 is retained. ProDistill uses64+64 unlabeled calibration images and official coefficient kernels. These are merger evaluations, not additional recovery attacks.

Table 8: Eight participating experts: three evaluated tasks

Protected task	Protection	No recovery M / D / S	AGA target M / D / S
MNIST	Unprotected	94.00 / 52.18 / 74.42	93.23 / 52.34 / 73.90
MNIST	Defense 1	9.80 / 5.85 / 6.70	9.80 / 5.90 / 6.70
MNIST	O/B [H]	51.73 / 50.64 / 40.57	54.22 / 50.80 / 44.63
MNIST	A/B [H]	61.94 / 50.21 / 53.09	59.16 / 50.59 / 51.97
MNIST	PaRaMS (full)	9.82 / 2.13 / 9.65	93.23 / 52.34 / 73.90
MNIST	PaRaMS attention ablation	94.00 / 52.13 / 74.45	93.23 / 52.34 / 73.90
DTD	Unprotected	94.00 / 52.18 / 74.42	94.06 / 51.38 / 74.46
DTD	Defense 1	72.22 / 2.13 / 52.48	72.30 / 2.13 / 52.47
SVHN	Unprotected	94.00 / 52.18 / 74.42	93.78 / 52.18 / 72.53
SVHN	Defense 1	9.80 / 27.07 / 6.70	9.80 / 27.07 / 6.70

TA0.2 with eight experts; cells always MNIST / DTD / SVHN, including partner tasks. Other five task datasets were not evaluated. No eight-task average. [H] Original task-bound failures retained. Complete PaRaMS and its Attention ablation remain distinct.

Table 9: Paper-formula Unmask: a separate eight-expert recovery setting

Protected task	Protection	No recovery M / D / S	Unmask0.3 M / D / S
MNIST	Unprotected	94.00 / 52.18 / 74.42	7.52 / 1.97 / 9.55
MNIST	Defense 1	9.80 / 5.85 / 6.70	10.28 / 2.07 / 7.76
MNIST	O/B [H]	51.73 / 50.64 / 40.57	11.35 / 1.97 / 17.59
MNIST	A/B [H]	61.94 / 50.21 / 53.09	10.89 / 2.13 / 12.85
MNIST	PaRaMS (full)	9.82 / 2.13 / 9.65	9.58 / 1.70 / 14.45
MNIST	PaRaMS attention ablation	94.00 / 52.13 / 74.45	7.51 / 1.97 / 9.55
DTD	Unprotected	94.00 / 52.18 / 74.42	10.00 / 2.55 / 9.56
DTD	Defense 1	72.22 / 2.13 / 52.48	10.28 / 2.13 / 7.76
SVHN	Unprotected	94.00 / 52.18 / 74.42	7.91 / 2.13 / 9.15
SVHN	Defense 1	9.80 / 27.07 / 6.70	11.35 / 2.23 / 19.59

TA0.2 with eight experts; cells always MNIST / DTD / SVHN, including partner tasks. Other five task datasets were not evaluated. No eight-task average. [H] Original task-bound failures retained. Complete PaRaMS and its Attention ablation remain distinct. Fixed merged parameters minus0.3 times the whole published expert. Both protected and unprotected controls receive the operation. Paper-formula native adaptation, not official code or a complete paper table. No AGA composition. The unprotected model also collapses; low absolute scores do not demonstrate successful expert recovery.

Table 10: Targeted bias cleanup: numerical protocols kept separate

Target / partner	Numerics	Scope	Control T / P	Defense 1 T / P
MNIST / SVHN	Native FP32	Target	99.37 / 92.50	99.37 / 92.50
MNIST / SVHN	Native FP32	Both	99.37 / 92.50	99.37 / 92.50
MNIST / DTD	Native FP32	Target	99.45 / 69.26	99.45 / 69.26
MNIST / DTD	Native FP32	Both	99.45 / 69.26	99.45 / 69.26
DTD / MNIST	FP32 prefix; FP64 readout	Target	69.26 / 99.45	69.26 / 99.45
DTD / MNIST	FP32 prefix; FP64 readout	Both	69.26 / 99.45	69.26 / 99.45
SVHN / MNIST	FP32 prefix; FP64 readout	Target	92.50 / 99.37	92.50 / 99.37
SVHN / MNIST	FP32 prefix; FP64 readout	Both	92.50 / 99.37	92.50 / 99.37

TA0.5; no images, original expert or secret perturbation enters recovery construction. The FP64-readout rows are a separately frozen protocol and must not be pooled with native FP32. Original native DTD control failure0.00082493 and MNIST before/after score difference0.00069427 remain recorded. Equal recovered predictions do not establish exact FP32 function identity.

Table 11: Task-direction changes and actual classification effects

Fixed model	Task-inside distance	Task-outside distance	Standalone M	TA M	AGA both M
Global permutation [K]	2.69895	11.86048	99.69	7.39	10.09
Previous compensation [F]	0.49964	11.85437	99.69	99.13	99.04
Distance-only pairing	2.96985	12.02151	99.69	13.99	11.08
Task-directed pairing	2.70843	12.03349	99.69	10.98	9.80

Same 74 matrices and original expert-minus-base rank 32 right singular directions. Squared per-matrix relative norms are summed. These matrix-only distances exclude biases/LN/embeddings and are not the full-model distance. Comparable task-outside distances do not imply comparable protection. [K] Known symmetry; [F] failed efficacy. Distance-only pairing is a fixed control,not a replacement selected by accuracy; – means no classification run. The task-directed release is selected before images; no novelty claim for standard symmetry/SVD/greedy alone.

Table 12: Fixed ablation of the task-direction objective

Construction	Standalone M	TA M / S	AGA target M / S	AGA both M / S
Task-directed	99.69	10.98 / 14.32	9.80 / 9.81	9.80 / 9.83
Distance-only	99.69	13.99 / 8.78	11.27 / 16.32	11.08 / 16.30

Same target, base, rank, 74 matrices, greedy rule, TA0.5 and original AGA. The distance-only order was fixed before task-directed images; its classification run was registered after the main MNIST result. All scores are retained without selecting a replacement. This isolates the objective in this fixed family; it is not a seed study or universal security claim.

Table 13: Previous Defense 2 candidate: source-only ablations

Counterfactual	Correct	Prediction changes	Max. relative gap change
Task-inside matrices only	511/512	1	1.05992
Task-outside matrices only	503/512	9	1.82071
All changed matrices	505/512	7	1.87001
Other parameters only	505/512	7	1.85216
Complete joint change	512/512	0	3.02985e-06

The fixed512 target-training source images, not a test table. Matrix inside/outside components use the same original rank32 task basis; non-matrix changes are separate. Each counterfactual is quantized separately. Complete-release replay and independent NumPy recount both have zero error. No candidate was repaired or selected from these counterfactual outcomes.

Table 14: Completed work and remaining comparisons

Item	Status
TA + original AGA	Completed: two-expert pairs; target/both recovery scopes; protected and partner accuracy.
Defense 1	MNIST, DTD, SVHN completed; targeted cleanup and fixed-budget hard-label recovery completed with separate permissions/numerics.
Defense 2	Fixed structure and the frozen task-SVD-directed construction verified on MNIST/DTD/SVHN. Parameter-only distance ablation retained. Method contribution and known-family recovery remain separate; not a complete eight-task benchmark.
PaRaMS	Full-default protection completed for MNIST/DTD/SVHN, including matched TA and original AGA. Attention-only ablation completed for MNIST; diagonal recovery is a native adaptation.
MergeLock	Not run: official repository currently contains only README/Coming Soon. No invented implementation or paper scores.
MergeGuard / ARF	Full defenses not run; training-stage conditions differ. Unmask formula was tested separately; GradErase not reproduced.
MergeBarrier	Full defense not run; architecture adaptation and verified implementation missing.
Eight-task benchmark	Eight experts participated; protected/evaluated targets cover MNIST/DTD/SVHN. Complete protected cases and evaluation for SUN397/Cars/RESISC45/EuroSAT/GTSRB are missing.
Other mergers	TIES, three DARE masks, NUFILT and ProDistill completed in documented scopes. Full TSV-Merge, CAT and LOT not run.
Reliability	Per-class/pair counts and source audits retained. Only DARE has the stated three-mask comparison; do not present deterministic rows as multi-seed results.

This is a completion inventory, not an accuracy table. Missing methods are not assigned zero accuracy. More merger names do not increase the number of recovery attacks.

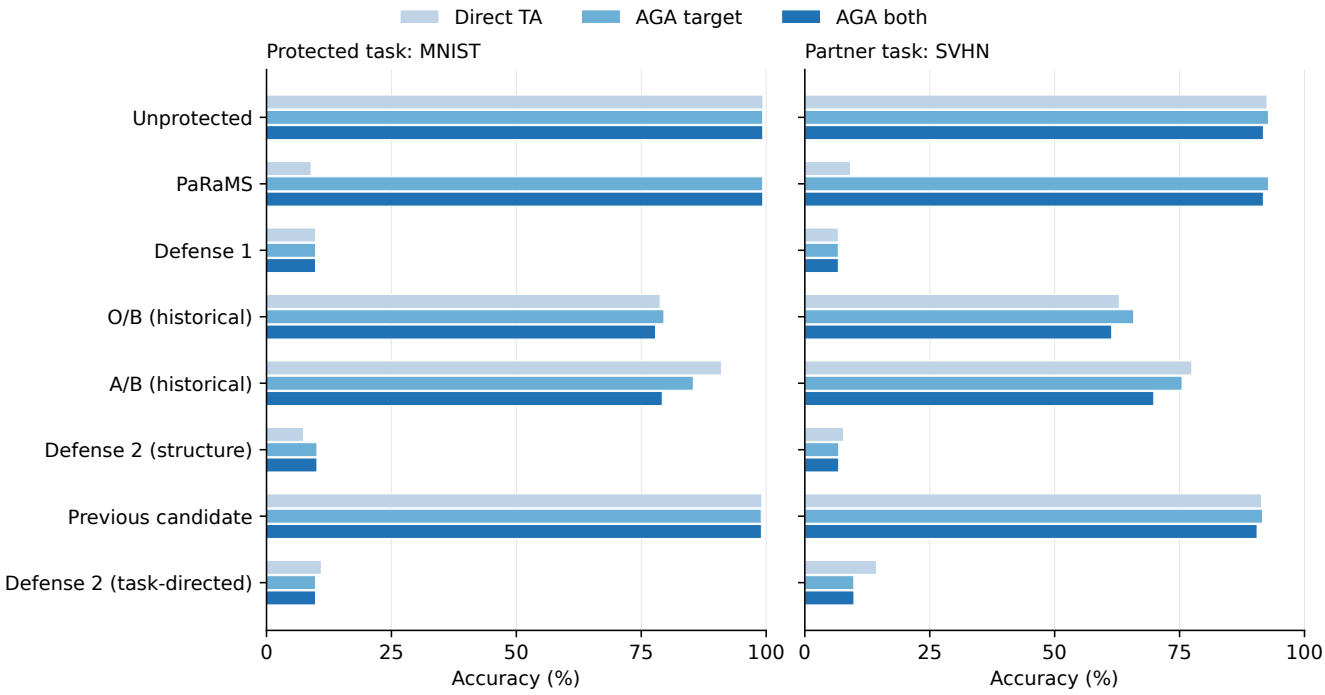

Figure 1: Two-expert TA0.5 before and after original AGA. Both protected and partner tasks are shown. Every bar is one fixed configuration; historical O/B and A/B task-bound failures remain labelled. Exact numbers appear in Table 1.